

27-8 Yahrzeit Server

The Yahrzeit Wall is an etched-glass memorial display with individually controlled LEDs. A dedicated server calculates Yahrzeit and Yizkor observances according to synagogue policy and sends the required lighting pattern to the wall controller.

Synagogue staff normally review and maintain the system through its browser interface. Lighting operates automatically, with manual controls available when needed.

Manufacturer: On-Site Systems Inc., Yyz Systems Inc, & Allan Schwartz

Model Number: *custom installation*

Maintenance: *See the Maintenance section in the User Guide.*

Service: Electronics – Yyz Systems
Software – Allan Schwartz
Repair/reprint panels of glass – On-Site Systems

Location: Sanctuary west wall

Installers:	On-Site Systems Inc.	http://www.on-sitesystems.com	
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	Yyz Systems Inc	http://www.yyzsystems.com	
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This section describes the Yahrzeit Server and its operator interface. See **section 27-6, Yahrzeit Overview**, for the complete system and **section 27-7, Yahrzeit Panels**, for the etched-glass panels.

The Yahrzeit system has three software components:

- **Web interface** — maintains memorial records and synagogue policy, displays status, and runs reports.
- **Automation software** — schedules Yahrzeit and Yizkor operations and sends lighting commands to the controller.
- **Controller firmware** — receives those commands and drives the Wall's LED array.

A dedicated Linux appliance in the synagogue office hosts the web interface and automation software. It communicates with the Arduino-based Yahrzeit Controller over the synagogue Ethernet network.

Accessing the Yahrzeit Server's User Interface

Synagogue staff and the Rabbi use the browser interface for routine operation. In any current web browser, open:

<http://192.168.2.148/yahrzeit>

Complete operating instructions are maintained in the online **User Guide**, available through the **User Guide** button in the Yahrzeit Wall interface.

Note: If the address above no longer works, contact the person responsible for the synagogue network or consult **section 27-4, Networking**. Record any permanent address change in this manual.

Logging in to the Yahrzeit Server via SSH

Routine operation is performed through the browser interface. SSH access is reserved for a technical administrator performing installation, recovery, or diagnostic work.

From a terminal, enter:

```
ssh <administrator>@<Yahrzeit-server-address>
```

Use the appliance administrator's login name and the same server address used for the browser interface. Administrator credentials and recovery information should be kept in the synagogue's secured technical records. Do not record a password in this House Manual.

Installing the Yahrzeit Software or Replacing the Yahrzeit Server

Replacing the Yahrzeit Server is a technical maintenance operation for a person familiar with Linux, networking, and command-line administration. The source code, installer, and current technical documentation are maintained at:

<https://github.com/allanschwartz/yahrzeit>

Begin with an Intel NUC or similar computer running a current Ubuntu Server LTS release. Connect it to the wired network and enable OpenSSH during Ubuntu installation.

Log in as the appliance administrator and run:

```
curl -fsSL \
```

```
https://raw.githubusercontent.com/allanschwartz/yahrzeit/master/  
yahrzeit\_site-v3/bin/install-yahrzeit.sh \  
-o /tmp/install-yahrzeit.sh
```

```
bash /tmp/install-yahrzeit.sh
```

The installer installs the required Linux packages, downloads the Yahrzeit application, configures Apache and PHP, enables SSH, and performs non-transmitting software checks. By default, it installs the application at:

```
~/src/yahrzeit/yahrzeit_site-v3
```

and makes the browser interface available at:

```
http://<appliance-address>/yahrzeit/
```

After installation, the technical administrator must:

1. Restore or verify [data/yahrzeits-rev4.csv](#).
2. Open the Minhag page, verify the synagogue settings, and save them.
3. Verify the Yahrzeit Controller address recorded by the installer.
4. Run the database audit and command preview.
5. Verify the browser interface before performing a live lighting test.

From the application directory, the initial checks are:

```
bin/yahrzeit --audit  
bin/yahrzeit --dry-run
```

Do not perform a live lighting operation until the controller address and the generated lighting pattern have been verified.

Additional technical instructions are located in:

```
yahrzeit\_site-v3/README.md  
yahrzeit\_site-v3/docs/yahrzeit-installer-notes.txt
```

Online User Guide

The online User Guide is reproduced after this section. The current version is also available through the **User Guide** button in the Yahrzeit Wall web interface.



Yahrzeit Wall User Guide

Operator guide for the Congregation Beth Sholom Yahrzeit Wall application.

Yahrzeit Wall User Guide

This guide explains the Congregation Beth Sholom Yahrzeit Wall web application at an operator level. It is intended for office staff, ritual committee members, and technical volunteers who maintain the memorial database, review reports, check wall assignments, or perform occasional manual wall operations.

What This Application Does

The Yahrzeit Wall application maintains the memorial-name database, applies Congregation Beth Sholom lighting policy, generates reports, audits memorial locations, and sends command streams to the embedded wall controller.

In normal operation, no one needs to manually light the wall each day. The scheduled automation decides when lighting should change and uses the same command path available from the web application and command line.

System Overview

The system has three main parts:

- **Yahrzeit Wall** — the physical memorial wall, panels, names, and LEDs.
- **Yahrzeit web application** — the PHP application that maintains data, reports, settings, and operator screens.
- **Embedded controller** — the Arduino-based controller that receives commands and drives the LED wall.

The normal scheduled-control path is:

```

cron
-> bin/yahrzeit_scheduler
    -> bin/yahrzeit
        -> bin/yahrzeit_engine.php
            -> controller command stream
        -> nc
            -> Yahrzeit controller (Arduino)

```

In practical terms: cron decides when scheduled phases run, `yahrzeit_scheduler` decides whether the requested phase applies today, `yahrzeit` decides how to run and transmit the action, and `yahrzeit_engine.php` decides what names should be lit, audited, or reported.

Normal Operation

Normal yahrzeit and Yizkor lighting is automatic. Cron runs a small number of named scheduler phases at the times saved on the Minhag page. Each scheduler phase either performs one action or does nothing.

11:00 AM `yahrzeit_scheduler --phase yizkor-on`
If today is a configured Yizkor day, turn on Yizkor lighting.

1:00 PM `yahrzeit_scheduler --phase yizkor-off`
If today is a configured Yizkor day, restore normal yahrzeit lighting.

Every evening at the configured fixed or sunset-relative time:
`yahrzeit_scheduler --phase yahrzeit`

The application maintains its own marked block in the appliance crontab. Normal Yahrzeit lighting is reapplied every evening. Fixed times remain stable. Sunset-based times are recalculated automatically as sunset changes through the year.

The web application is mainly used for review, maintenance, reporting, auditing, and occasional manual operation. Routine daily lighting should not require manual web-page action.

Common Tasks

To find a memorial name, use the [Names page](#). To see where a person is assigned on the wall, open the [memorial record](#) or view the [relevant panel](#).

To review upcoming Yahrzeit observances, use the [Reports page](#). Day, week, next-week, month, and next-month reports list names whose yahrzeits fall in the selected period. The next enabled Yizkor dates are shown on the [Yizkor page](#) and in the [Minhag preview](#).

To check the integrity of the memorial database, use the [Reports page](#) to run the audit. The audit checks for unknown panel IDs, invalid row or column values, malformed records, and duplicate assignments to the same LED location.

To preview what would be sent to the controller, use the command preview on the [Reports page](#). Preview does not transmit commands to the wall.

To replace the memorial database, use the CSV upload tool on the [Reports page](#). The current CSV file is backed up before replacement, and an audit is run after upload.

To change synagogue-wide lighting policy, use the [Minhag page](#). These settings affect the entire wall and should match Congregation Beth Sholom practice.

Page-by-Page Guide

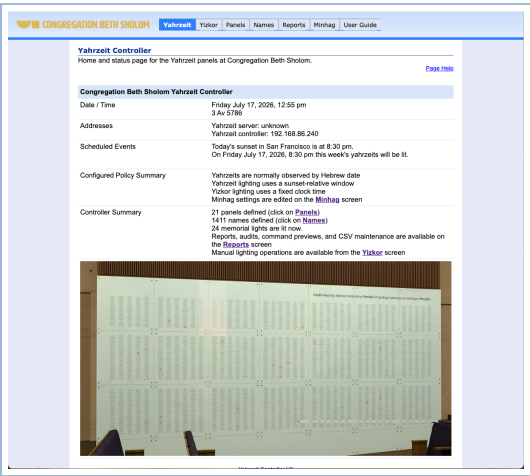
Each page has its own help link. The descriptions below explain what each page shows and when to use it.

[Yahrzeit Wall Home](#)

The Yahrzeit Home page provides an at-a-glance summary of the appliance and its current operating context. It shows the civil and Hebrew dates, the server and controller addresses, the next scheduled lighting event, and a summary of the saved lighting policy.

It also reports the number of configured panels and memorial records, and how many Yahrzeit lights should currently be lit. Use this page as the starting point when checking the system or confirming its configuration. Links in the summary lead to the appropriate maintenance pages.

See also [Yahrzeit Wall Home Help](#).



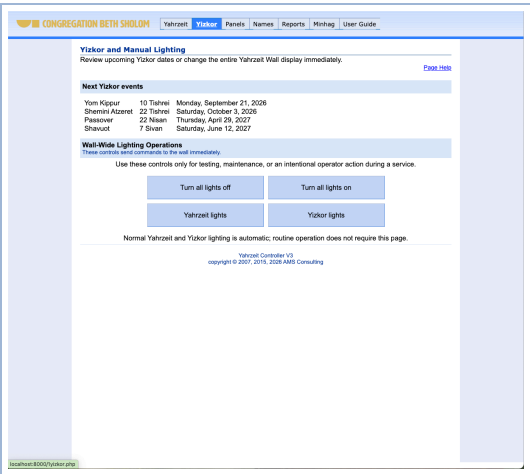
Yizkor and Manual Lighting

The Yizkor page lists the next enabled Yizkor observances, in upcoming date order. The dates reflect the Yom Kippur, Shemini Atzeret, Pesach, Shavuot, and optional additional observances selected on the Minhag page.

This page also provides immediate control of the complete Yahrzeit Wall. **Turn all lights off** and **Turn all lights on** are intended for testing and maintenance. **Yahrzeit lights** restores the normal date-based display, while **Yizkor lights** turns on the full-wall memorial display.

Normal Yahrzeit and Yizkor lighting is automatic, so these controls are not ordinarily needed. They may also be used for an intentional operator action during a service.

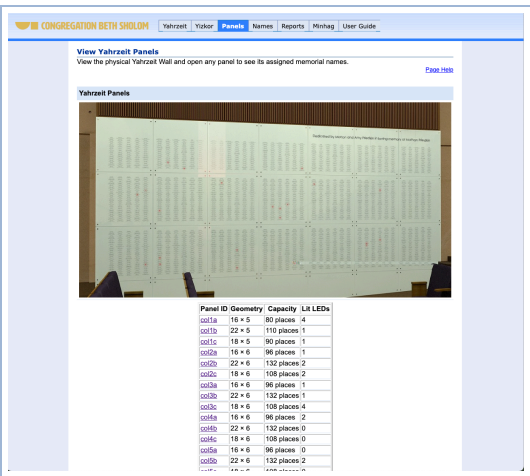
See also [Yizkor and Manual Lighting Help](#).



View Yahrzeit Panels

The Panels page shows the physical layout of the Congregation Beth Shalom Yahrzeit Wall. Each glass panel in the photograph can be selected; click a panel in the photograph to open a detailed view of the memorial names assigned to it.

The table below the photograph provides another way to select a panel. It lists each panel ID, its row-and-column geometry, its total memorial capacity, and the number of LEDs that should currently be lit. Click a panel ID to open the same detailed panel view.



Use this page to locate a panel, review the overall wall configuration, or begin checking database assignments against the physical wall. The panel geometry represents the installed wall and is not edited from this page.

See also [View Yahrzeit Panels Help](#).

Single Yahrzeit Panel

The Single Panel page shows the memorial names assigned to one physical glass panel of the Congregation Beth Sholom Yahrzeit Wall. Its grid follows the installed rows and columns, so each screen position corresponds to one possible memorial-light position on the panel.

Occupied positions show the assigned memorial name and an LED symbol indicating whether that memorial light should currently be lit. Empty positions remain open. The summary below the grid gives the total number of memorial lights that should be lit on the selected panel.



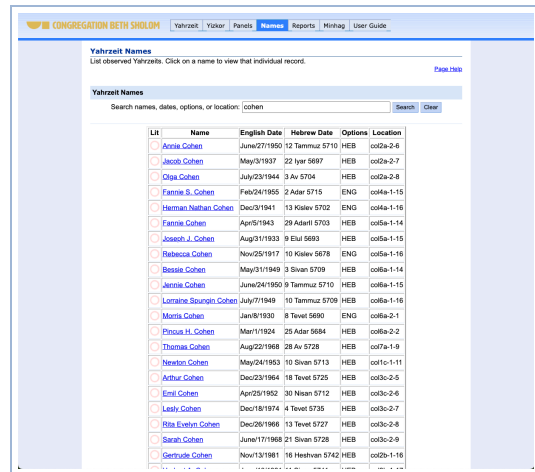
Use this view to compare database assignments with the names etched on the physical glass panel. Click any memorial name to open that individual record for closer review or correction. If a position appears duplicated or invalid, run the database audit from the Reports page.

See also [Single Panel Help](#).

Yahrzeit Names

The Names page is the main viewer for the memorial database. It lists all memorial records and provides a search field for finding people by name, English or Hebrew date, observance option, or panel location. Searching is not case-sensitive.

More than one search term may be entered. A record is displayed only when all the terms are found, even if they occur in different fields. For example, a surname may be combined with a year, Hebrew month, or panel ID to narrow the results. Select **Clear** to display all memorial records again.



Each result shows the memorial name, stored English and Hebrew dates, special observance options, and assigned panel location. The Lit symbol indicates whether the memorial light should currently be lit. Click a person's name to open the individual memorial record for review or correction.

New memorial names have historically been added in batches, rather than one at a time from this page. Use the [Reports page](#) to download the current CSV database, edit the downloaded file while preserving its CSV format, and upload the replacement file. Records are removed through the same controlled CSV-replacement process. The upload operation first backs up the current database and then audits the replacement.

See also [View Names Help](#).

Single Yahrzeit Name

The Single Name page displays one individual memorial record and permits careful corrections to that record. It shows the person's name, stored English and Hebrew dates, and the panel, column, and row assigned to the memorial light.

The observance choices specify whether the family follows the English or Hebrew Yahrzeit date and whether additional memorial observances apply. A normal record uses automatic, calendar-driven lighting. A location reserved for future use may instead be marked **Reserved**, which keeps its light off.

Use this page for individual corrections, such as fixing a name, date, or panel location. After selecting **Save**, the application backs up the current memorial database, replaces only the selected record, and automatically runs the database audit. Review any reported audit problem before making another change.

See also [Memorial Record Help](#).

Yahrzeit Reports

The Reports page generates lists of memorial names whose Yahrzeits fall on a selected day, during this or next week, or during this or next month. The anchor date determines which day, week, or month is selected.

Week reports use the Erev Shabbat-to-Erev Shabbat observance period. The displayed civil dates run from Saturday through Friday. When the anchor date is Friday, **This week** begins that evening, while **Next week** begins the following Friday evening.

The diagnostic tools check the relationship between the memorial database and the physical wall. **Audit Database** reports invalid panel IDs, row or column values outside the panel geometry, malformed records, and duplicate LED assignments. **Preview Commands** shows the controller commands that would be generated today without transmitting them to the wall.

The Reports page also maintains the complete CSV memorial database. **Download CSV** saves a copy outside the appliance. **Upload CSV** installs a replacement database for batch maintenance. Before replacement, the current database is backed up; after replacement, the new database is automatically audited.

Example — Next Week: With Friday, July 17, 2026 as the anchor date, this report selects the following observance week: Saturday, July 25 through Friday, July 31. Each row gives the observance date, memorial name, Hebrew and English dates, wall location, and date option.

CONGREGATION BETH SHOLOM

Yahrzeit

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Yahrzeit Reports

Generate Yahrzeit reports, preview controller commands, and export or import the Yahrzeit database.

Page Help

Report: next-week for 2026-07-17 (exit 0)

Yahrzeit report: next-week
Range: Saturday July 25, 2026 through Friday July 31, 2026
Count: 30

Date	Name	Hebrew Date	English Date	Location	Options
2026-07-25	Boris Milovich	11 Av 5731	Aug/2/1971	col2b-4-5	HEB
2026-07-25	Isaac Schutz	11 Av 5728	Aug/5/1968	col2c-5-14	HEB
2026-07-25	Joseph William Hymson	11 Av 5708	Aug/16/1948	col1c-3-3	HEB
2026-07-25	Louis Kinspel	11 Av 5681	Aug/15/1921	col1a-2-10	HEB
2026-07-25	Ruth Zepora Katz	11 Av 5703	Aug/12/1943	col2c-3-7	HEB
2026-07-25	Sarah Weinberg	11 Av 5722	Aug/11/1962	col3c-6-15	HEB
2026-07-25	Yetta Leah Steinhof	11 Av		col5a-6-4	HEB
2026-07-26	Capt. Harry H. Morris	12 Av 5704	Aug/1/1944	col2a-4-15	HEB
2026-07-26	Ethel Schwartz	12 Av 5698	Aug/9/1938	col6a-6-1	HEB
2026-07-26	Martin Braverman	12 Av 5699	July/28/1939	col4a-1-8	HEB
2026-07-26	Necha Katznelson	12 Av 5732	July/23/1972	col1b-3-6	HEB
2026-07-26	Necha Katznelson	12 Av 5732	July/23/1972	col3b-3-13	HEB
2026-07-26	Ruth B. Garter	12 Av 5760	Aug/13/2000	col5b-1-16	HEB
2026-07-28	Cilla Kirsch	14 Av 5717	Aug/11/1957	col4a-3-14	HEB
2026-07-28	Samuel Martyn	14 Av 5754	July/22/1994	col7a-3-8	HEB
2026-07-29	Annie Litman	15 Av		col3a-4-10	HEB
2026-07-29	Arthur N. Spivock	15 Av 5693	Aug/7/1933	col1a-5-10	HEB
2026-07-29	Clara Forman	15 Av 5721	July/28/1961	col4a-2-14	HEB
2026-07-29	Esther Palonsky	15 Av 5717	Aug/12/1957	col2a-5-5	HEB
2026-07-29	Henry Hinden	15 Av 5737	July/30/1977	col3c-3-13	HEB
2026-07-29	Johanna Mayerhoff	15 Av 5729	July/30/1969	col2b-4-3	HEB
2026-07-30	Lily C. Levy	16 Av 5740	July/29/1980	col1b-3-15	HEB
2026-07-30	Rose Barrett	16 Av 5725	Aug/14/1965	col6a-1-3	HEB
2026-07-30	Zelda Korss Davis	16 Av 5741	Aug/16/1981	col1c-2-1	HEB
2026-07-31	Frank S. Nathan	17 Av 5697	July/25/1937	col1c-4-9	HEB
2026-07-31	Joseph Marcus	17 Av 5674	Aug/9/1914	col6a-4-13	HEB
2026-07-31	Rose Morris Mund	17 Av 5730	Aug/19/1970	col2b-4-7	HEB
2026-07-31	Sally Slotnick	17 Av 5726	Aug/3/1966	col3c-6-3	HEB
2026-07-31	Viola B. Shenson	17 Av 5728	Aug/11/1968	col3c-5-18	HEB
2026-07-31	Yhoshva Zimmerman	17 Av 5712	Aug/8/1952	col1c-5-18	HEB

Return to Reports

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See also [Reports Help](#).

Yahrzeit Minhag

The Minhag page records synagogue-wide Yahrzeit and Yizkor practices. It also stores the synagogue name and affiliation displayed by the application. Because these settings affect the entire wall, they should be changed only when the intended synagogue practice is understood.

The Yahrzeit settings choose whether regular observances follow the English or Hebrew date, whether lighting changes at a fixed time or a selected number of minutes before sunset, and whether each Yahrzeit is observed for its day only or for the full Erev Shabbat-to-Erev Shabbat week.

The Yizkor settings enable the four annual Yizkor observances and select the appropriate Pesach and Shavuot days. An optional additional Hebrew date may be configured for another full-wall memorial observance. The preview lists the next civil date of every currently saved observance. Yizkor on/off times may use fixed clock times or a window around sunset.

CONGREGATION BETH SHOLOM

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Yahrzeit Minhag

Special minhag or customs used for Yahrzeit Wall specific to Congregation Beth Shalom. These observances apply uniformly to all individuals.

Edit Data

Yahrzeit Application Customization for your Synagogue

Synagogue Name: Congregation Beth Shalom

Synagogue Affiliation: Conservative

Yahrzeit Calendar Driven Automation: Can be overridden for individuals

observe the English date

observe the Hebrew date

Yahrzeit Scheduled Run Time: Run at 05:00 pm 18 minutes before sunset

Lighting Option: Choose whether each Yahrzeit is lit for its observed day or for the full Shabbat-to-Shabbat week

Observe the Yahrzeit day only

Observe the full week, from Erev Shabbat through the following Erev Shabbat

Synagogue Yizkor Minhag

Yizkor dates observed: Up to five memorial dates may be selected. These extra observances apply to all individuals

Yom Kippur

Shmini Atzeret

1st day of Pesach

2nd day of Shavuot

Other date

Near Yizkor events: Based on the settings currently saved

Yom Kippur: 10 Tishrei Monday, September 21, 2026

Shmini Atzeret: 22 Tishrei Saturday, October 3, 2026

Pesach: 22 Nisan Thursday, April 29, 2027

Shavuot: 7 Sivan Saturday, June 12, 2027

Yizkor Light On/Off Times: Set time On at 05:00 pm Off at 05:30 pm 18 min. before Sunset to 72 min. after sunset

SAVE

Yahrzeit Controller V3
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Selecting **Save** writes the revised settings and regenerates the appliance's managed lighting schedule. The result page displays the installed schedule or reports that schedule installation failed. After changing observance dates or lighting times, review the preview and resulting schedule before relying on automatic operation.

See also [Minhag Settings Help](#).

Safety Notes

Some operations affect the live wall or the live database. Use extra care with these actions:

- **Manual wall-wide lighting controls** send commands to the controller immediately.
- **CSV upload** replaces the live memorial database.
- **Minhag changes** affect scheduling and lighting policy for the whole wall.

After changing memorial data, run the audit before relying on scheduled lighting. After changing Yizkor or timing settings, review reports or previews as appropriate to confirm the intended behavior.

Manual Wall Operations

The Yizkor page provides manual wall-wide operations: all on, all off, the normal Yahrzeit display, and full-wall Yizkor lighting. These controls are useful for testing, maintenance, or special operator action. They are not normally required for daily scheduled operation.

Manual restoration may be useful after testing, maintenance, a power interruption, or an incorrect wall display. After using **Turn all lights on**, **Turn all lights off**, or **Yizkor lights**, select **Yahrzeit lights** to recalculate and restore the normal date-based lighting pattern.

Because these controls act immediately, use them only when intentionally changing the current wall display. See the [Yizkor page](#) for these operations.

Database and Files

The live memorial database is stored as a CSV file:

data/yahrzeits-rev4.csv

The synagogue lighting-policy settings are stored in:

data/minhag.ini

Normal web pages read these files through the application include files rather than parsing them directly. The Reports page provides the safest web path for CSV download, upload, audit, and preview.

Modifying the Memorial Database

Correcting one memorial record:

1. Open the [Names page](#) and locate the memorial using the search field.
2. Select the person's name, make the required correction on the Single Yahrzeit Name page, and select **Save**.
3. Review the audit result shown after the record is saved. The application creates a timestamped backup before replacing the record.

4. Open the affected Single Panel page and compare the displayed name and position carefully with the actual etched-glass panel.

Adding records or making batch changes:

1. On the [Reports page](#), select **Download CSV** and retain the downloaded file as a backup.
2. Make the required changes to a working copy of the CSV file. Preserve the existing columns, field meanings, and panel-location format.
3. Select **Upload CSV** and choose the corrected file.
4. Review the audit results displayed after the upload. If errors are reported, correct the working copy and upload it again.
5. Review every affected panel on the Single Panel page and compare it with the corresponding etched-glass panel.

The server backs up the existing live database before installing an uploaded replacement. That local backup does not replace the external downloaded copy described below.

Backup

Use **Download CSV** on the Reports page to save a copy of the current memorial database. Make a backup after significant database changes and before performing a batch update. Add the date to the downloaded filename and keep the copy somewhere other than on the Yahrzeit appliance.

The application also makes a timestamped local backup before replacing the database during an upload. These local copies provide protection against an incorrect upload, but they do not replace an off-appliance backup that remains available if the appliance or its storage fails.

Restore

To restore the memorial database, use **Choose File** on the Reports page to select a known-good backup, and then select **Upload CSV**. The application backs up the current database before replacing it and automatically audits the restored database.

Review the audit results before relying on reports or scheduled lighting. If the audit reports errors, correct the CSV file or restore a different backup and upload it again.

Maintenance

The Yahrzeit Wall requires little routine physical maintenance. While the wall is operating correctly, no internal maintenance of the LED boards, interconnect cables, or controller is recommended.

Routine maintenance consists primarily of:

- Keeping the Yahrzeit appliance, Yahrzeit Controller, and wall power supplies continuously powered and connected to the wired network.
- Maintaining a recent external copy of the memorial database.
- Reviewing the Home page and `data/automation.log` after an appliance replacement, operating-system update, network change, or power interruption.
- Testing the Yizkor pattern before a Yizkor observance. Confirm that the complete wall illuminates, and then select **Yahrzeit lights** to restore the normal display.

The appliance calculates the Yahrzeit and Yizkor dates automatically. The schedule does not require annual entry of calendar dates.

Technical Overview

The web application and embedded controller are intentionally layered. On the site side, the scheduler decides when to act, the wrapper decides how to run and transmit, and the engine decides what names should be lit or reported.

On the embedded controller side, the network or serial input code collects command lines, the command processor parses them, the wall abstraction applies logical row/column/panel operations, and the low-level pixel driver updates the hardware.

```
socket_thread / serial_thread
  -> CmdProc
    -> LedWall
      -> YyzPixel
```

Troubleshooting

If the wall does not display the expected pattern, check the system in this order:

1. Confirm power to the Yahrzeit appliance, Yahrzeit Controller, and wall power supplies.
2. Confirm that the appliance and controller are connected to the synagogue Ethernet network.
3. Open the browser interface. Review the Home page, the expected lighting pattern, the memorial database, and the Minhag settings.
4. Run the database audit from the Reports page and review `data/automation.log`.
5. Have a technical administrator verify communication between the appliance and the Yahrzeit Controller.

If only one LED or a localized group of LEDs fails while the rest of the wall operates normally, the likely cause is a physical LED, pixel-board, or interconnect-cable fault. Investigate the physical wall only after power, software, database, and controller communication have been verified.

If a report looks wrong, confirm its anchor date and check the Minhag settings that control the date method and observance period. If a name appears in the wrong wall position, run the database audit, correct the memorial record or CSV database, and audit again.

When in Doubt

For ordinary review, use Names, Panels, and Reports. For live-wall changes, CSV upload, or Minhag changes, proceed deliberately and run audit or preview checks afterward.